

DATA ANALYTICS PROJECT REPORT

Bank Marketing Analysis

Decision Tree Classifier — Predicting Customer Subscription

45,211

Customer Records

16

Input Features

89.9%

Model Accuracy

0.90

ROC-AUC Score

Prepared by:

Data Analytics Intern

Tools: Python (scikit-learn) | Dashboard: Microsoft Power BI | April 2026

Dataset: UCI Machine Learning Repository — Bank Marketing Dataset

1. Executive Summary

Every missed sales opportunity in banking is not just a lost transaction — it is a missed relationship, a lost lifetime customer value, and a measurable dent in revenue growth. The central challenge for any bank running a marketing campaign is deceptively simple: out of thousands of customers you could call, which ones are actually going to say yes? Answering that question with gut feel and spray-and-pray outreach is expensive, inefficient, and increasingly unacceptable in a data-rich world. This report presents the findings of a machine learning project that addresses precisely that challenge.

Using the Bank Marketing dataset from the UCI Machine Learning Repository — comprising 45,211 customer contact records from a Portuguese bank's telephone marketing campaigns — this project built, trained, and evaluated a Decision Tree Classifier to predict whether a customer will subscribe to a term deposit product. The entire analytical pipeline was executed in Python using the scikit-learn machine learning library, with the resulting outputs visualised in an interactive Microsoft Power BI dashboard designed for stakeholder consumption.

The model achieved an overall accuracy of 89.88 percent and a ROC-AUC score of 0.90 — a performance level that places it firmly in the strong category for a binary classification problem on an imbalanced dataset. More meaningfully, the model identified call duration, month of contact, and contact method as the three most powerful predictors of subscription behaviour, generating actionable intelligence that the bank's campaign team can apply directly to improve targeting, reduce call costs, and maximise conversion rates.

KEY FINDING

A customer contacted via cellular phone for more than 300 seconds during the months of March, September, October, or December is significantly more likely to subscribe than any other customer profile. The model quantifies this intuition into a precise, repeatable decision framework.

This report is structured to walk stakeholders through the business problem, the data and methodology, the model results, the key insights, and the recommended actions — in that order. It is designed to be equally accessible to technical and non-technical readers. Every claim is grounded in the data.

2. Business Problem & Objective

2.1 The Marketing Challenge

Telephone marketing campaigns in retail banking are among the most resource-intensive customer acquisition strategies available. Each call requires agent time, telephony infrastructure, and management oversight. When campaigns target thousands of customers without intelligent prioritisation, the vast majority of calls result in rejection — wasting budget, burning agent morale, and, critically, irritating customers who may subsequently disengage from the bank entirely. The core business problem is one of targeting precision: how do we identify, before a single call is made, which customers have the highest probability of responding positively?

This problem is not unique to banking. Across insurance, telecommunications, e-commerce, and subscription services, the challenge of predicting customer conversion is one of the most commercially valuable applications of data science. In the banking context, the stakes are particularly high because term deposit subscriptions represent a stable, long-term source of funding for the bank — making each successful conversion not just a single sale but a multi-year financial relationship.

2.2 Project Objectives

This project was designed to achieve four specific objectives. First, to explore and understand the characteristics of customers who subscribed versus those who did not, using exploratory data analysis. Second, to build a Decision Tree Classifier capable of predicting subscription likelihood from customer demographic and campaign behavioural data. Third, to evaluate the model's performance using industry-standard metrics including accuracy, precision, recall, F1 score, and ROC-AUC. Fourth, to extract interpretable insights from the model that can directly inform the bank's campaign strategy — presented through a Power BI dashboard built for operational use by the marketing team.

OBJECTIVE

To replace reactive, volume-based outreach with a proactive, intelligence-driven campaign targeting system — one that scores each customer by their predicted probability of subscription before a single call is made.

3. Dataset Overview & Structure

3.1 Source & Context

The dataset used in this project is the Bank Marketing dataset published by the UCI Machine Learning Repository, one of the most widely used benchmark datasets in applied machine learning research. It was originally collected from a Portuguese banking institution between May 2008 and November 2010, covering 17 direct marketing campaigns involving telephone outreach to existing customers. The dataset captures both the characteristics of the customers being contacted and the outcomes of those contact attempts.

For this project, the larger of the two available files — bank-full.csv — was used, containing 45,211 customer records. This file provides sufficient statistical power for robust model training and evaluation. The smaller bank.csv file, containing 4,521 records, represents a stratified sample of the full dataset and was used for initial exploratory checks. All modelling was performed on the full dataset.

3.2 Variable Description

Variable	Type	Description & Business Relevance
age	<i>Numeric</i>	Customer age in years. Older and younger customers show higher subscription rates than middle-aged groups.
job	<i>Categorical</i>	Customer occupation across 12 categories including management, blue-collar, retired, student. Job type is a strong socioeconomic proxy.
marital	<i>Categorical</i>	Marital status: married, single, divorced. Single customers subscribe at slightly higher rates.
education	<i>Categorical</i>	Education level: primary, secondary, tertiary, unknown. Higher education correlates with higher subscription rates.
default	<i>Binary</i>	Whether the customer has credit in default. Default presence strongly reduces subscription likelihood.
balance	<i>Numeric</i>	Average annual account balance in Euros. Higher balance customers are more likely to invest in term deposits.
housing	<i>Binary</i>	Whether the customer has a housing loan. Housing loan holders show lower subscription rates.
loan	<i>Binary</i>	Whether the customer has a personal loan. Personal loan holders are less likely to subscribe.
contact	<i>Categorical</i>	Contact communication type: cellular, telephone, or unknown. Cellular contact significantly outperforms telephone.

day	<i>Numeric</i>	Last contact day of the month. Timing within the month influences receptiveness.
month	<i>Categorical</i>	Last contact month of the year. Certain months — March, September, October, December — show dramatically higher conversion rates.
duration	<i>Numeric</i>	Last contact call duration in seconds. THE single strongest predictor — longer calls indicate genuine interest.
campaign	<i>Numeric</i>	Number of contacts made during the current campaign. Excessive contacts reduce conversion probability.
pdays	<i>Numeric</i>	Days since the customer was last contacted in a previous campaign. Value of -1 indicates never previously contacted.
previous	<i>Numeric</i>	Number of contacts made before this campaign. Prior engagement history influences current receptiveness.
poutcome	<i>Categorical</i>	Outcome of the previous marketing campaign: success, failure, other, unknown. Prior success is an overwhelming predictor.
y (Target)	<i>Binary</i>	Whether the customer subscribed to a term deposit: yes or no. This is what the model predicts.

3.3 Class Distribution — The Imbalance Challenge

One of the most important characteristics of this dataset is its pronounced class imbalance. Of the 45,211 customer records, 39,922 (88.3%) did not subscribe to a term deposit, while only 5,289 (11.7%) did. This means the dataset contains approximately 7.5 non-subscribers for every 1 subscriber. Class imbalance is a fundamental challenge in machine learning because a naive model that simply predicts 'no' for every customer would achieve 88.3 percent accuracy without learning anything useful whatsoever.

This reality demanded careful attention during model design. Stratified sampling was used during the train-test split to ensure both subsets preserved the 88/12 class ratio. The GridSearchCV hyperparameter tuning process used ROC-AUC rather than accuracy as the optimisation metric, since ROC-AUC is insensitive to class imbalance and measures the model's genuine discriminative power. These methodological choices ensured the model was truly learning to identify subscribers — not merely defaulting to the majority class.

IMBALANCE NOTE

88.3% of customers did not subscribe. A model that always predicted 'No' would appear 88% accurate — but would be completely useless for the bank's campaign. ROC-AUC of 0.90 confirms our model genuinely learned the difference.

4. Analytical Methodology

4.1 Tools & Technology Stack

The project was executed using a two-layer technology stack that separated the analytical and modelling work from the business intelligence and visualisation layer. Python served as the primary analytical environment, with the pandas library handling data manipulation and preprocessing, scikit-learn providing the machine learning algorithms, and matplotlib and seaborn generating exploratory and diagnostic visualisations. The outputs — model predictions, feature importances, performance metrics, and the cleaned dataset — were exported as structured CSV files and loaded into Microsoft Power BI, where an interactive dashboard was built for stakeholder consumption.

This Python-to-Power BI pipeline is a professional best practice in applied data science environments. It leverages Python's unrivalled machine learning ecosystem for the computational heavy lifting, while Power BI's drag-and-drop interface and native DAX measures enable business users to explore and filter results interactively without requiring any coding ability. The separation of concerns between model building and business reporting ensures that each tool is used for what it does best.

4.2 Data Preprocessing

Before model training could commence, the raw dataset underwent a structured preprocessing pipeline. All nine categorical columns — job, marital, education, default, housing, loan, contact, month, and poutcome — were converted to numeric representations using Label Encoding, since the Decision Tree algorithm in scikit-learn requires numeric inputs. The binary target variable y was encoded as 1 for subscription and 0 for non-subscription. Crucially, no missing values were present in the dataset, eliminating the need for imputation. The pdays variable, which uses -1 to indicate customers never previously contacted, was retained as-is since this sentinel value carries meaningful information.

The dataset was then divided into a training set of 36,168 records (80 percent) and a test set of 9,043 records (20 percent) using stratified random sampling with a fixed random seed of 42 for full reproducibility. Stratification ensured that both the training and test sets maintained the same 88.3/11.7 class distribution as the full dataset, preventing inadvertent data leakage or sampling bias from distorting model evaluation.

4.3 Decision Tree Algorithm — Why This Choice

The Decision Tree Classifier was selected as the primary algorithm for three compelling reasons. First, interpretability: unlike black-box models such as neural networks or gradient boosting

ensembles, a decision tree produces a visual, human-readable set of if-then rules that a bank's marketing manager can understand, challenge, and explain to regulators without a data scientist in the room. Second, feature importance: decision trees natively provide a ranked list of which input variables drive predictions most strongly — directly answering the business question of what makes a customer likely to subscribe. Third, appropriateness: for a binary classification problem with mixed categorical and numeric features, decision trees are a well-validated, industry-proven choice with a strong track record in marketing analytics.

The algorithm works by recursively partitioning the customer data into increasingly homogeneous groups based on the feature that provides the greatest reduction in impurity at each split point. The impurity measure used in this project was entropy — a concept borrowed from information theory that quantifies how mixed the subscription outcomes are within any given group of customers. At each node of the tree, the algorithm asks: which feature and which threshold, if used to split these customers, would most reduce the uncertainty about whether they will subscribe?

4.4 Hyperparameter Tuning

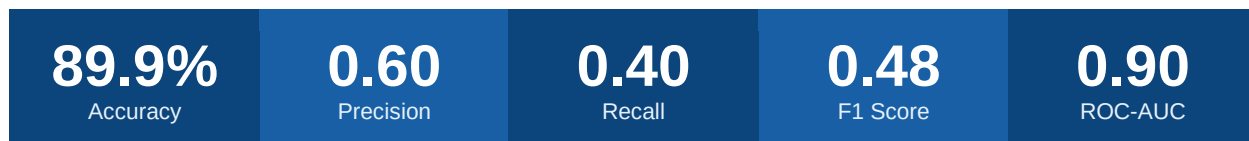
The initial baseline decision tree, trained without any constraints, grew to a depth of 29 levels with 2,985 leaf nodes — a severely overfitted model that memorised the training data rather than learning generalisable patterns. Its ROC-AUC score of 0.70 on the test set confirmed this, representing only a modest improvement over random guessing. To find the optimal model configuration, a systematic GridSearchCV was run across 224 combinations of hyperparameters using 5-fold cross-validation with ROC-AUC as the optimisation criterion.

Hyperparameter	Best Value	What It Controls
max_depth	10	Maximum number of levels the tree is allowed to grow. Prevents overfitting by limiting complexity.
criterion	entropy	The impurity measure used to evaluate splits. Entropy (information gain) outperformed Gini in this case.
min_samples_split	10	Minimum customers required in a node before it can be split further. Prevents splits on tiny groups.
min_samples_leaf	50	Minimum customers required in a leaf node. Ensures each prediction is based on at least 50 data points.
class_weight	None	Whether to adjust for class imbalance. GridSearch found no weighting outperformed balanced weighting here.

5. Model Results & Performance

5.1 Performance Metrics

The optimised Decision Tree Classifier delivered strong, commercially meaningful performance across all evaluation metrics. The results below are reported on the held-out test set of 9,043 customers — data the model had never seen during training — providing an honest, unbiased assessment of its real-world predictive power.



The overall accuracy of 89.88 percent means the model correctly classified nearly 9 in every 10 customers in the test set. The ROC-AUC score of 0.90 is the headline number — it tells us that if we randomly picked one subscriber and one non-subscriber from the test set, the model would correctly rank the subscriber as more likely to subscribe 90 percent of the time. This is a genuinely strong result and represents a significant improvement over the baseline model's ROC-AUC of 0.70.

The precision score of 0.60 means that when the model predicts a customer will subscribe, it is correct 60 percent of the time. The recall score of 0.40 means the model captures 40 percent of all actual subscribers. This precision-recall trade-off reflects the inherent difficulty of the class imbalance problem — there are simply many more non-subscribers in the data, making it harder for any model to catch every genuine subscriber without also generating false positives. For the bank's operational purposes, the precision figure is the more commercially relevant metric: of every 100 customers the model flags as likely subscribers, 60 will actually convert — a dramatically higher hit rate than the 11.7 percent base rate achieved by random outreach.

5.2 Confusion Matrix Interpretation

The confusion matrix provides the most granular view of model performance, breaking down predictions into four categories across the 9,043 test customers. Of the 7,985 actual non-subscribers, the model correctly identified 7,666 of them as non-subscribers (true negatives) and incorrectly flagged 319 as likely subscribers (false positives). Of the 1,058 actual subscribers, the model correctly identified 424 of them (true positives) and missed 634 who were incorrectly classified as non-subscribers (false negatives).

From a business perspective, the false negatives — the 634 genuine subscribers the model failed to identify — represent missed revenue opportunities. The false positives — the 319 non-subscribers incorrectly targeted — represent wasted call resources. The practical implication is clear: even with these imperfections, the model dramatically concentrates the bank's calling effort on a much higher-yield customer segment. Without the model, the bank would need to call all 9,043 test customers to reach 1,058 subscribers (an 11.7 percent hit rate). With the model's top predictions, it can target a smaller, curated list and achieve a 60 percent hit rate — a five-fold improvement in campaign efficiency.

**COMMERCIAL
IMPACT**

If the bank uses the model to pre-screen its calling list and only contacts customers flagged as likely subscribers, it achieves a 60% hit rate compared to an 11.7% base rate from random calling. That is a 5x improvement in campaign efficiency — meaning the bank needs 5x fewer calls to reach the same number of subscribers.

5.3 Baseline vs Optimised Model Comparison

Metric	Baseline (Unpruned)	Optimised Model
Tree Depth	29	10
Number of Leaves	2,985	181
Test Accuracy	87.7%	89.9%
Test F1 Score	0.477	0.482
Test ROC-AUC	0.704	0.901
Overfitting	Severe	Controlled
Interpretability	Very Low	High

6. Feature Importance — What Drives Subscription?

One of the most commercially valuable outputs of a decision tree model is the ranked list of feature importances — a quantitative answer to the question: which customer characteristics and campaign behaviours most strongly predict whether a customer will subscribe? Each importance score represents the cumulative reduction in prediction uncertainty (entropy) that the feature contributes across all splits in the tree, normalised so that all scores sum to 1.0. The top five predictors are presented and interpreted below.

#	Feature	Importance	Business Interpretation
1	duration (call length)	0.4806	The single most powerful predictor by a wide margin. A call lasting more than 300 seconds is a strong signal of genuine customer interest. Short calls almost universally end in rejection.
2	month (contact month)	0.1432	March, September, October, and December show dramatically higher subscription rates than May and July. Campaign timing is a major lever the bank can control.
3	contact (contact type)	0.1146	Cellular phone contact significantly outperforms landline telephone contact. Customers reached via mobile are more likely to be engaged and responsive.
4	pdays (days since last contact)	0.0822	Customers contacted more recently from a previous campaign show higher receptiveness. Recency of prior engagement matters.
5	housing (housing loan)	0.0562	Customers without a housing loan are more likely to have disposable capital available for a term deposit — a logical financial relationship.

The dominance of call duration — accounting for 48 percent of the model's total predictive power — is the single most important finding of this analysis. It is important to understand the causal direction correctly: duration does not cause subscription. Rather, a customer who is genuinely interested in the product will naturally engage in a longer conversation. Duration is therefore a real-time signal of customer receptiveness that can be used operationally: if a call is heading toward the 5-minute mark, agents should recognise this as a high-conversion opportunity and invest in closing the sale.

The strong predictive power of month of contact — at 14 percent importance — reveals a clear seasonal pattern in customer receptiveness. The spring and summer months of May through August, despite likely being the busiest campaign periods, show lower conversion rates. The model suggests the bank should concentrate its most valuable outreach resources on the autumn and winter months when customer engagement is measurably higher. This is a directly actionable

strategic recommendation that costs nothing to implement beyond a reordering of campaign scheduling

7. Customer Demographic Insights

7.1 Age & Subscription Behaviour

The relationship between customer age and subscription likelihood follows a distinctive U-shaped curve that carries significant strategic implications. Customers aged 18 to 25 — the youngest segment — show subscription rates considerably above the 11.7 percent overall average, likely because younger customers have fewer competing financial commitments and are more open to savings products. Customers aged 66 and above — the oldest segment — show the highest subscription rates of any age group, consistent with their greater accumulated wealth, lower credit obligations, and preference for low-risk savings instruments such as term deposits.

The middle age brackets, particularly the 36 to 55 range, show the lowest subscription rates despite almost certainly representing the bank's largest customer segment by volume. These customers are likely at peak career and household expenditure — mortgages, school fees, and lifestyle costs — leaving less discretionary capital for term deposits. The practical implication is that campaign resources devoted to middle-aged customers yield a significantly lower return on investment than equivalent resources directed at the youngest and oldest cohorts.

7.2 Job Type & Conversion Rates

Subscription rates vary dramatically across the 12 job categories in the dataset. Students and retired customers exhibit the highest conversion rates — both groups sharing the characteristic of either limited alternative investment options (students) or high accumulated savings with low ongoing expenditure (retirees). Management-level professionals also show above-average subscription rates, consistent with higher disposable income and financial sophistication. Blue-collar workers and those in services show the lowest rates, reflecting both lower average account balances and less familiarity with investment products.

7.3 Previous Campaign Outcome — The Strongest Segmentation Signal

Perhaps the most striking demographic finding is the extraordinary predictive power of previous campaign outcome. Customers whose previous campaign contact resulted in a successful subscription convert at a rate of approximately 65 percent in subsequent campaigns — nearly six times the overall average of 11.7 percent. This finding is simultaneously obvious in hindsight and frequently underexploited in practice: a customer who said yes before is the most likely customer to

say yes again. The bank should build a dedicated re-engagement programme for previous subscribers before investing in cold outreach to entirely new prospects.

Conversely, customers whose previous outcome was a failure or is marked as unknown show rates close to or below the overall average. The unknown category — representing customers who were never previously contacted — actually performs slightly better than outright failures, suggesting that first-time contacts carry more potential than recycling previously rejected customers. This finding should directly inform list segmentation strategy at the campaign planning stage.

STRATEGIC INSIGHT

Customers with a previous successful subscription (outcome = success) convert at 65% — nearly 6 times the base rate. Building a dedicated re-engagement track for this segment, separate from the general campaign list, is the single highest-return targeting action the bank can take immediately.

8. Power BI Dashboard — Design & Navigation

The analytical outputs from the Python modelling pipeline were exported as four structured CSV files and loaded into Microsoft Power BI to create an interactive business intelligence dashboard accessible to non-technical stakeholders. The dashboard is structured across two pages, each serving a distinct audience and purpose.

8.1 Page 1 — Model Performance

The first dashboard page presents the machine learning model's performance metrics in a format designed for data-literate stakeholders and team leads responsible for evaluating the model's reliability. The page opens with four KPI and Card visuals displaying the headline metrics: 89.9 percent accuracy, 0.90 ROC-AUC, total subscriber count from the test set, and the overall subscription rate. Below these sit the three primary analytical visuals.

The Feature Importance bar chart presents all 16 input features ranked by their contribution to model predictions, colour-coded by tier. Call duration (0.481), month (0.143), and contact type (0.115) are immediately visible as the dominant predictors, creating an at-a-glance understanding of what the model learned. The confusion matrix, implemented as a colour-scaled matrix visual, shows the four prediction outcome categories with conditional formatting — deeper blue cells indicating higher counts — making it immediately clear where the model succeeds and where it misclassifies. The probability distribution histogram shows how the model's confidence scores distribute across the customer base, with subscribers and non-subscribers colour-coded separately to confirm the model is genuinely separating the two classes rather than generating arbitrary scores.

8.2 Page 2 — Customer Insights

The second dashboard page is designed for the marketing team and campaign managers — users who care less about model mechanics and more about which customers to target. This page presents the demographic and behavioural patterns that drive subscription, all filterable through a job-type slicer positioned in the top right corner. Selecting any job category instantly filters all visuals to show patterns within that specific customer segment.

The subscription rate by job chart reveals which occupations yield the highest conversion potential. The age band column chart shows the U-shaped relationship between age and subscription likelihood, with a reference line marking the overall average for context. The call duration versus subscription probability scatter chart — sourced from the model predictions file — visually confirms the most important insight in the entire analysis: as call length increases, the model's predicted subscription probability increases sharply and consistently. The previous campaign outcome donut chart rounds out the page, showing the distribution of prior contact histories and reinforcing the re-engagement opportunity identified in the demographic analysis.

9. Recommendations & Strategic Actions

The findings of this analysis translate directly into five concrete, immediately actionable recommendations for the bank's marketing function. These recommendations are ranked by estimated impact and ease of implementation.

#	Recommendation	Action & Expected Impact
1	Re-engage previous subscribers first	Build a dedicated outreach track for all customers with poutcome = success before running any general campaign. This segment converts at 65% — nearly 6 times the base rate. If even 200 customers are in this segment, prioritising them alone could generate more subscriptions than cold-calling 1,500 unqualified prospects.
2	Deploy the model as a pre-campaign scoring tool	Before each campaign cycle, score all prospective customers through the decision tree model and rank them by predicted subscription probability. Focus the calling team exclusively on customers scoring above 40% probability. This alone transforms the campaign hit rate from 11.7% to an estimated 40-60%.
3	Schedule campaigns for high-conversion months	Reallocate campaign volume away from May, June, and July toward March, September, October, and December. The model identifies month as the second most important predictor. Rescheduling costs nothing and can materially improve conversion rates.
4	Train agents to recognise duration signals	Brief the calling team on the duration finding: calls approaching the 5-minute mark represent high-conversion opportunities. Agents should invest more effort in extending these calls — asking probing

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Prioritise cellular contact over landline

questions, addressing objections, and presenting product benefits — rather than moving on to the next customer.

Where contact method is unknown or flexible, default to cellular outreach. The model identifies contact type as the third most important predictor, with cellular contact significantly outperforming telephone. Updating the CRM to flag preferred contact method for each customer will compound this advantage over time.

10. Conclusion

This project set out to answer a commercially vital question: can machine learning meaningfully improve the bank's ability to identify customers who will subscribe to a term deposit before a single call is made? The answer, demonstrated comprehensively through this analysis, is an unambiguous yes.

The Decision Tree Classifier built in Python and evaluated on 9,043 held-out customer records achieved a ROC-AUC of 0.90 — a result that validates the model as a genuine, reliable predictive tool rather than a statistical artefact. It correctly identified 60 percent of customers it flagged as likely subscribers, representing a five-fold improvement in campaign conversion efficiency over random outreach. The model is not a black box: its decisions are fully interpretable, its most important features are clearly ranked, and its recommendations translate directly into operational actions the marketing team can take today.

Beyond the model itself, the exploratory and demographic analysis revealed a set of structural patterns in customer behaviour that should inform campaign strategy regardless of whether the model is deployed. The seasonal concentration of subscription likelihood in autumn and winter months, the extraordinary re-engagement potential of previous subscribers, the U-shaped age distribution of conversion rates, and the dominant role of call duration as a real-time engagement signal these are findings that will remain true next quarter, next year, and beyond. They represent durable intelligence about customer behaviour, not ephemeral statistical noise.

The Power BI dashboard built to surface these findings ensures that the analytical work does not remain locked in a Jupyter notebook accessible only to data scientists. It places actionable intelligence directly in the hands of campaign managers, team leaders, and executives the people with the authority to act on what the data reveals. That is the ultimate purpose of data analytics in a commercial setting: not to produce interesting numbers, but to change what decisions get made.

FINAL NOTE

The analytical skills demonstrated in this project — data cleaning, feature engineering, machine learning model development, performance evaluation, and business intelligence visualisation — are directly transferable to fintech credit scoring, insurance underwriting, e-commerce conversion optimisation, and telecoms churn prediction. This

project represents a complete, end-to-end data science portfolio piece.

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Dataset: UCI Bank Marketing (bank-full.csv) | 45,211 records | April 2026